

PROJECT SPECIFIC RAM PLAN - DSTW -
DIGITAL FIELD ELEMENTS

Solution Element

Digital Field Elements

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LOG OF CHANGES			
Revision	Date	Author	Change Description
01 PD 01	2025-06-13	Florian Moser	First proposal
01 RL	2025-06-25	Florian Moser	First edition
02 PD 01	2025-09-24	Florian Moser	See section 1.5.3
02 RL	2025-10-03	Florian Moser	See section 1.5.4

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1. SCOPE

1.1 PURPOSE

This document presents the Reliability, Availability and Maintainability Plan (RAM) of the solution element Digital Field Elements (DFE) Phase 1.

The RAM plan defines the reliability, availability and maintainability objectives, tasks to be performed and documents to be produced and delivered at various phases of the project's lifecycle, to address the functional and technical requirements specified in the respective feature lists and requirements documentation with main focus on hardware.

For information about organisation, work packages, milestones and application of the Management System refer to [SEMP_DFE].

1.2 IDENTIFICATION

Project:	DSTW ÖBB Entwicklung
Solution:	100025795 / ATRELDE91
Solution / System:	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
Solution Element:	Digital Field Elements (DFE) Phase 1 ¹
Identification:	DFE 1.0.0 IFC 1.0.0
Contract:	Digitales Stellwerk DSTW – Rahmenvereinbarung
Customer:	ÖBB
End User:	ÖBB
SIL Level	SIL 4

Table 1: Solution Identification

1.3 INTENDED AUDIENCE

The intended audience of the project specific RAM Plan are the following functions:

- Project Design Authority (PDA)
- Project Manager (PM)
- Solution Engineering Manager (SEM)
- RAM Assurance Manager (RAM-AM)

¹ For information about the different phases of DFE refer to [SEMP_DFE].

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- Safety Assurance Manager (SAM)
- Quality Assurance Manager (QAM)
- Validator (VAL)
- Verifier (VER)
- Security Assurance Manager (SecAM)

1.4 DOCUMENT MANAGEMENT

The management of this document will be performed according to the Document Management Rules [DMR] and the Document List and Review Matrix [DLRM].

1.5 CHANGE HISTORY

1.5.1 EDITION 01 PD 01

- First proposal.

1.5.2 EDITION 01 RL

- Review comments considered.

1.5.3 EDITION 02 PD 01

- Updated references.
- Changes due to updated [SEMP_DFE]:
 - X25 to X25oRaSTA Converter cancelled
 - Converter Zustimmung cancelled
 - EPD Box excluded
 - Retrofit cancelled
- Update of section 3.6 RAM Design Rules

1.5.4 EDITION 02 RL

- Review comments considered.

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

Id	Reference
[CRS_DSTW]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte; 3BU 17300 0004 DSAPC, Ed.04
[PH_RAMs]	Pflichtenheft Digital Interlocking OBB - D2.2 Teilpflichtenheft RAMS, 3BU 17300 0022 DTAPC, Ed.02
[RAM_Plan_Generic]	RAM Plan Generic; 3BU 15000 1712 DUAPA, Ed.06
[RAMP_DIL]	Project specific RAM Plan – DIL – Generic Application / Solution Element Virtual Interlocking; 3BU 17300 1209 DUAPA, Ed.01
[SEMP_DFE]	SEMP Solution Element - Digital Field Elements; 3BU 62800 0000 DPAPA, Ed.02
[PBS_DSTW]	Product Breakdown Structure (PBS) - Generic Application - Digital Interlocking OBB; 3BU 17300 1215 DPAPA, Ed.02
[SP_DSTW]	Safety Plan DSTW; 3BU 17300 1204 DUAPA, Ed.01
[IPP_DSTW]	IPP Digital Field Elements Phase 1; 3BU 62800 0050 MCAPA, Ed.01PD02

Table 2: Referenced Documents

2.2 STANDARDS AND PROCEDURES

Id	Reference
[Alcarel]	Reliability Standard Failure Rate Factor; 1AA 00210 0001 BCZZA
[Birolini]	Reliability Engineering. Theory and Practice; ISBN: 354040287X, 6 th Ed
[DLRM]	Document List and Review Matrix for engineering projects; 83530224-DOC-AUT-EN-010
[DMR]	Document Management Rules; 83530061-DOC-AUT-EN-013
[EN 50126-1]	EN 50126-1:2017 Railway applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS), Part 1: Generic RAMS process.
[IEC60050(192)]	IEC 60050-192:2015; International Electrotechnical Vocabulary (IEV) - Part 192: Dependability
[MIL_HDBK217F]	US MIL-HDBK 217F:1991, Reliability prediction for electronic equipment

Table 3: Standards and Procedures

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2.3 TERMS, ABBREVIATIONS AND ACRONYMS

Term/Abbreviation	Meaning
AzLM	Multi Section Axel Counter (one type of TDS)
DCA	Distributed Controller Architecture
DFE	Digital Field Elements
DIL	Digital Interlocking = DSTW
DSTW	Digitales Stellwerk = DIL
EKSA	Eisenbahnkreisungssicherungsanlage = LCSi (CFI ÖBB)
EULYNX	European Initiative to Linking Interlocking Systems
IEC	International Electrotechnical Commission
IFB	Interface Board
IFC	Interface Converter
Failure rate	Limit for $\Delta \rightarrow 0$ of the probability that an item will fail in the time interval $(t, t + \Delta]$, given that the item was new at $t = 0$ and did not fail in the interval $(0, t]$, divided by Δ . [Birolini]
Fault tree	A logic diagram showing which fault modes of subitems or external events, or combinations thereof, result in a given fault mode of the item. [IEC60050(192)]
FiT	Failures in time (unit: $10^{-9}h^{-1}$)
FES	Field Element Station
Ground Benign (GB)	Nonmobile, temperature and humidity-controlled environments readily accessible to maintenance; includes laboratory instruments and test equipment, medical electronic equipment, business and scientific computer complexes, and missiles and support equipment in ground silos. [MIL_HDBK217F].
Ground Fixed (GF)	Moderately controlled environments such as installation in permanent racks with adequate cooling air and possible installation in unheated buildings; includes permanent installations of air traffic control radars and communication facilities [MIL_HDBK217F].
[MIL_HDBK217F]	The Military Handbook is published by the US Department of Defense, and provides mathematical models for electrical, electronic and electro-mechanical components. Each model calculates the component failure rate for given combinations of environmental conditions, component value and type, electrical stress conditions and manufactured quality level
MTBF	The expectation of the operating time between failures. [IEC 60050(191)]. $MTBF$ is usually reserved for items with <i>constant</i> failure rate λ . In this case, $MTBF = 1/\lambda$ is the expected value (mean) of the <i>exponentially distributed item's failure-free operating time</i> . In particular $MTBF \approx T/k$, where T is the given, <i>fixed</i> cumulative operating time and k the total number of failures (failed items) during T . [Birolini]
MTTR	The expectation of the time to restoration. [IEC 60050(191)] $MTTR$ is the expected value (mean) of an item's repair time. It is obtained from the distribution function $G(t)$ of the repair time as $MTTR = \int_0^{\infty} (1 - G(t))dt$. For reliability or availability calculation a constant repair rate μ (i.e. exponentially distributed repair times with $\mu = 1/MTTR$) can often be used to get approximate results. [Birolini]
PBA	Printed Board Assembly

Table 4: Terms, Abbreviation and Acronyms

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3. DESCRIPTION OF THE SYSTEM

For detailed information on the products used, refer to [PBS_DSTW].

3.1 SYSTEM OVERVIEW

As stated in [SP_DSTW], the EULYNX system definition defines following Object Controller (OC) level subsystems:

- Subsystem – Point
- Subsystem – Light Signal
- Subsystem – Generic I/O
- Subsystem – Train Detection System (TDS)
- Subsystem – Level Crossing (LX)

Additionally, two interface converters are utilized:

- IFC ZG62 (Blockconverter)
- IFC Sbl60 (Blockconverter)

3.2 SOLUTION ELEMENTS

According to [IPP_DSTW], the Solution Element contains five work packages: (see also [SEMP_DFE])

- IFC ZG62 HW + SW
 - Based on ÖBB-Pilot DCA 2.x
 - Development/Integration of Computing Board SCA, DCA 2.x IOC, Relay Group
- FES M AzLM (TDS v7.10)
 - Development/Qualification of FES Modules
 - Module Network
 - Module Cable Distribution Frame
 - Module Power Supply
 - Integration AzLM ACE
- FES S DCA (FES S (up to 2 OC))
 - Development/Integration of the DCA 3.x (Object Controller Release 1 for EULYNX BL4R4 generic functionalities and OBB specific functionalities) version (HW)
 - Development/Qualification of FES Modules
 - Module Network
 - Module Cable Distribution Frame
 - Module Power Supply
 - Development/Qualification of additional Equipment
 - INDUSI-Extender (for Subsystem Signal)
 - CAN-Extension for Distances > 100m (for Subsystem Signal)
 - Development/Qualification of Subsystems
 - Point
 - Light Signal
 - Generic I/O (Station Key Lock, Line Key Lock, Local Control Post)
- FES M DCA
 - Same as FES S DCA, with extended capacity

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- FES S IFC LCSL (start of development in phase 1, finalization phase 2)
 - Support Supplier in LCSL Gateway development (Mission Embedded)

3.3 DCA 3.x COMPONENT VERSIONS

- DCA-P 3.1.0
 - Interface Control Unit Point - ICB-P HW 3.0
- DCA-LS 3.1.0
 - Interface Control Unit Output – ICB-O2
 - Interface Control Unit Generic IO – ICB-IO
- DCA-I/O 3.1.0
 - Interface Control Unit Input/Output High Power – ICB I/O HP
 - Interface Control Unit Input – ICB-I2
- ICB Carrier Boards
 - Carrier 19 Inch for 4 ICB
 - Carrier 19 inch for 2 ICB (Left)
 - Carrier 19 inch for 2 ICB

3.4 INTERFACE CONVERTER CABINETS

The IFC Cabinets include:

- IFC ZG62 1.0
 - Computing Board SCA
 - DCA 2.x IOC
 - Relay Group

3.5 FES CONFIGURATION

As per [SEMP_DFE]:

- Station Size S: Equipped exclusively with DCA 3.x units (LCSL Gateway in phase 2).
- Station Size M: Equipped with DCA 3.x units and AzLM units for TDS.
- Interface Converters (ZG62): Equipped with SCA and DCA 2.x for communication with neighboring interlockings

The next two figures (taken from [SEMP_DFE]) show a schematic buildup of a FES M DCA and a FES M AzLM.

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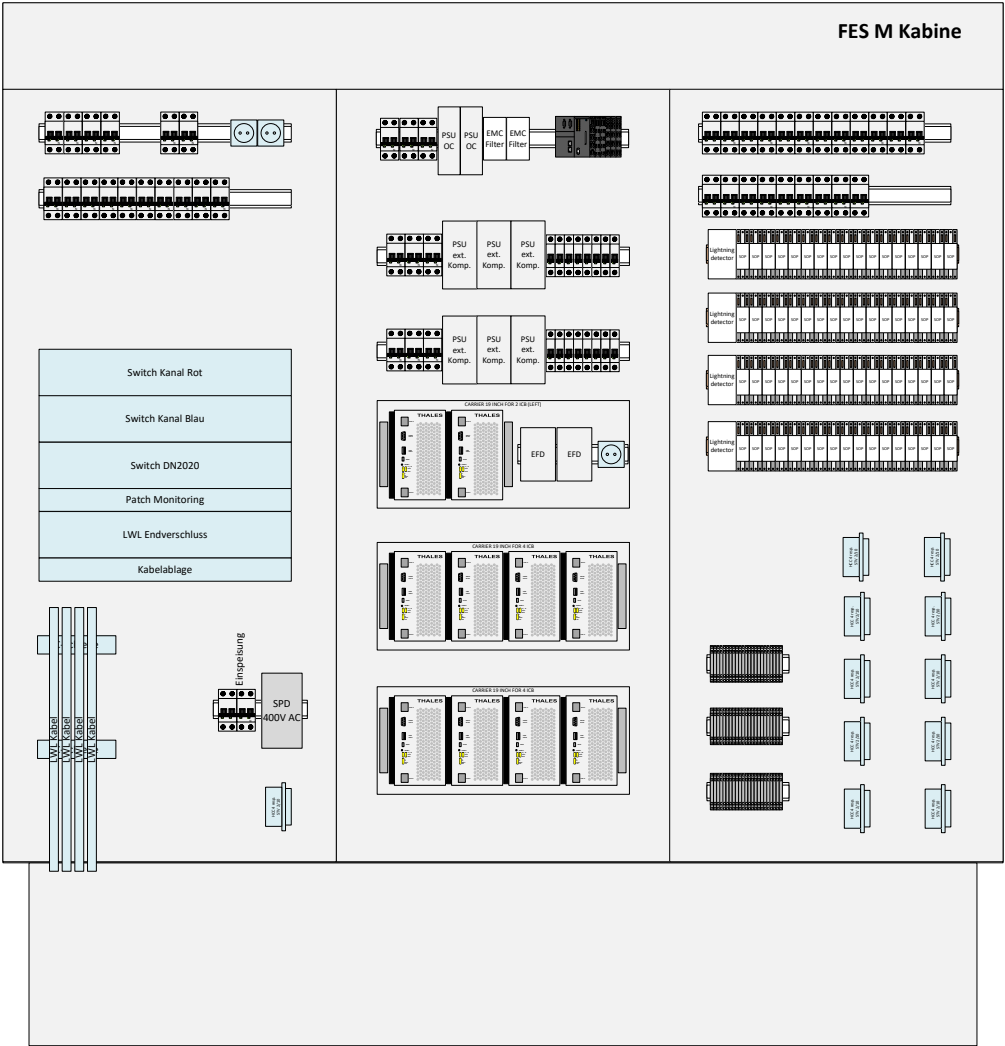


Figure 1: FES M DCA Schematic

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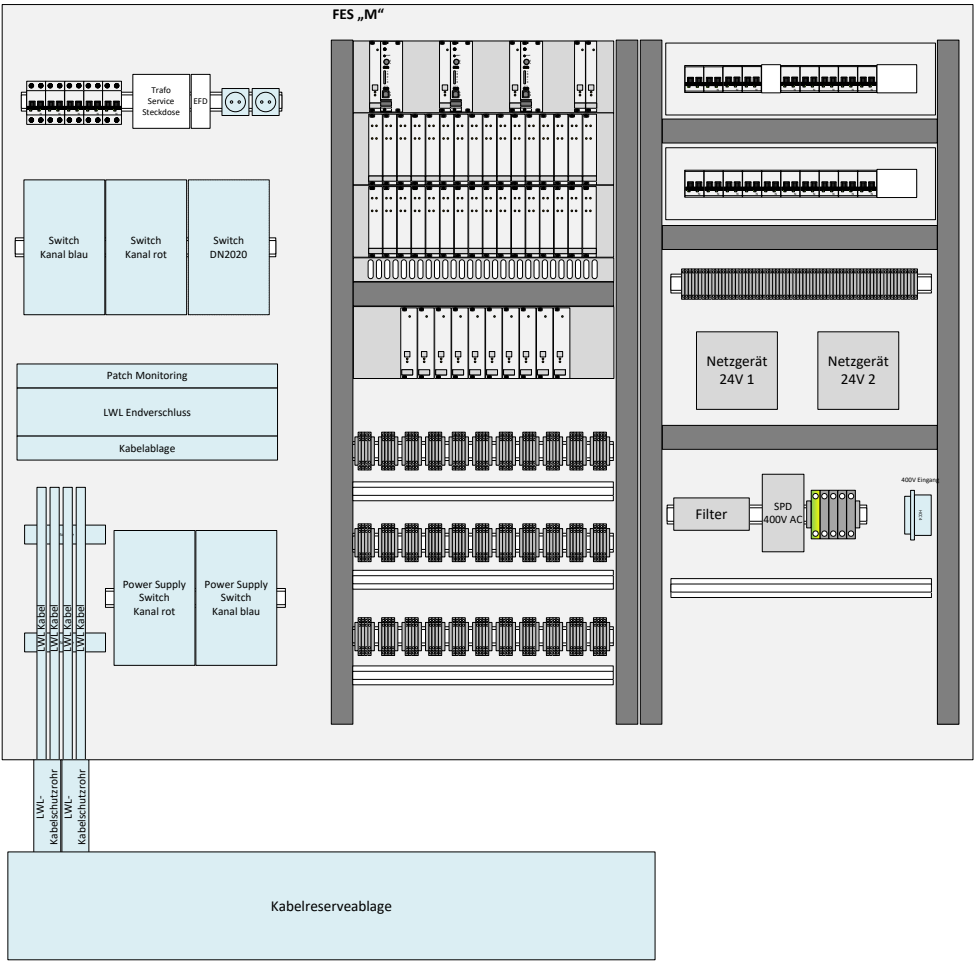


Figure 2: FES M AzLM Schematic

3.6 RAM DESIGN RULES

In order to ensure that the availability requirements (stated in section 6.2) are met, a series of rules, the so-called ‘RAM Design Rules’, will be developed and used as binding conditions for architectural design decisions. Note that each of these RAM Design Rules will also be **explicitly mapped to one or more specific system requirements**, ensuring traceability and compliance throughout the development lifecycle.

The common definitions and partitioning of the RAM Design Rules is described in detail in [RAMP_DIL] and is not in scope of this document. The following table presents the RAM Design Rules related to DFE.

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ID	Synopsis	Body
Group 01: General		
01.1	Diagnostic logging	All shutdown events of FES and OC components shall be provided in a unified customer-friendly way as well as for technical analyses. This information is then revised so that maintenance measures, for example, can be filtered. These data have to be made available, technically, and revised offline for the customer. Comment: also includes Blockconverter and EKSA
01.2	Reintegration	Reintegration of FES and OC components shall be provided in a unified customer-friendly way as well as for technical analyses. Comment: also includes Blockconverter and EKSA
01.3	Mapping component failure to severity level	The operative consequences of a failed hardware component shall be defined by the severity level of the availability class.
01.4	Mapping corrective maintenance action to severity level	The operative consequences during a corrective maintenance activity shall be defined by the severity level of the availability class.
01.5	Separate phase protection	Each power distribution unit (PDU) has to be separately protected. This means that an outage of one phase has no impact on other phases. Critical HW components which are related to availability must be supplied by separate phases.
01.6	Enable mapping of severity levels	The architectural concept shall be defined in a way to enable a mapping of severity levels to failed / reintegrated components.
Group 02: Availability		
02.1	OC Failure Isolation	A failure of one OC must not affect the operation of other OCs within the same FES.
02.2	Power Failure Resilience	FES must support automatic recovery after power restoration without manual intervention.
02.3	Power Supply Redundancy	Critical HW components which are related to availability must be equipped with two independent power supplies.
Group 03: Diagnostics		
03.1	Local Failure Indication	Each OC must provide local visual indicators (e.g., LEDs) for power, communication, and fault status.
03.2	Diagnostic Data Support for Maintainability	Ensure that Field Element Stations (FES) and Object Controllers (OC) provide structured and accessible diagnostic data to support efficient fault detection, analysis, and corrective maintenance. (Requires SDI-compliant diagnostic interface on all OCs). Comment: also includes Blockconverter and EKSA
Group 04: Maintainability		
09.1	Hot-Swap Capability	OCs must be replaceable without shutting down the FES or affecting other OCs.

Table 5: RAM Design Rules DFE

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3.7 AVAILABILITY CLASSES

[PH_RAMs] states the following availability classes for the Digital Interlocking. Each of the three reliability classes are subdivided into three subclasses addressing different MTTRs ("Mean Time To Restore") ranging from fast automatic failure correction to mean time for manual restoration. This breakdown is shown in Table 6 below.

Class	Availability [%]	MTTR [min]	MTBF [y]	TDT [min/y]
K1	99.999			5.25
		62.5	50	1.25
		10	5	2
		2	1	2
K2	99.995			26.25
		225	20	11.25
		10	2	5
		2	1	2
K3	99.99			52.5
		225	10	22.5
		10	2	5
		2	1	2

Table 6: Reliability figures by classes according to severity level

MTTR	Mean Time To Restore
MTBF	Mean Time Between Failures
TDT	Total Down Time

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4. RAM REQUIREMENTS

The reliability, availability and maintainability requirements have to be identified and referenced within the Dependability Analysis. This Dependability Analysis should also provide a proper apportionment and refinement of these requirements according to the system design. A general description of all RAM aspects can be found in [RAM_Plan_Generic].

4.1 RELIABILITY REQUIREMENTS

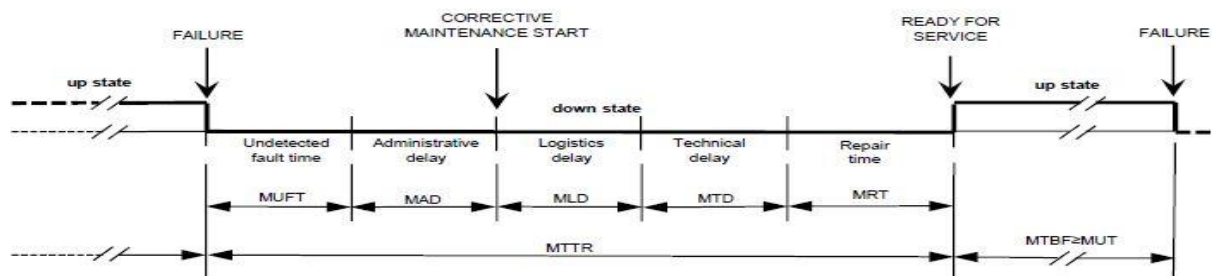
Concerning reliability, RAM activities have to focus on the following goals:

- Determination of all possible failures of the system with focus to random hardware faults. Analysis of their consequences with respect to reliability, availability and maintainability (MTTR) and calculation of the associated failure rates.
- Determination of the impact of failures on the functionality of the system.

The key parameters for describing RAM features of a system, on the basis of [EN 50126-1] are:

- Mean Up Time MUT, resp. Mean Time Between Failure MTBF,
- Mean time to Restore MTTR and the
- Availability A, resp. Unavailability ($U = 1 - A$).

Figure 3, obtained from [EN 50126-1], illustrates key parameters of the availability concept.



Key

MTBF	mean (operating) time between failures
MUT	mean up time
MUFT	mean undetected fault time
MAD	mean administrative delay
MLD	mean logistics delay
MTD	mean technical delay
MRT	mean repair time
MTTR	mean time to restore (for corrective maintenance)

Figure 3: Availability concept and related terms

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The MUT is the mean lifetime of a system between two failures. MTTR gives the time required from failure to restart operation.

4.2 AVAILABILITY REQUIREMENTS

In order to show an appropriate behaviour of the system and sub-systems with respect to availability, a system analysis shall be carried out to determine availability figures of components and sub-systems. The result shall be part of the Reliability Analysis. It shall be pointed out, that the availability figures depend on the proper installation, maintenance and operation conditions of the system and sub-systems as stated in the installation and maintenance manuals.

4.3 MAINTAINABILITY REQUIREMENTS

Concerning maintainability, RAM activities have to focus on the following goals:

- Identification and implementation of diagnostic measures to minimize system downtime
 - Detect fault alarms that have to be recorded during operation of the system.
 - Prepare appropriate counter measures in case of failures.
- Determine the time necessary for failure diagnosis and recovery of the availability of the system in case of system or component failures.
- Identification of necessary maintenance actions including time required to conduct them.

All requirements concerning maintenance activities need to be summarized to plan necessary actions. Those actions subsume inspection, preventive and corrective maintenance, repair and spare part logistic.

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5. RAM PLAN

5.1 METHODS AND TOOLS USED

The reliability calculation will be done according to [EN 50126-1] and the [MIL_HDBK217F] based Alcatel Reliability Standard [Alcarel] and according to the environmental conditions *Ground Benign (GB)* for indoor components and *Ground Fixed (GF)* for outdoor components.

5.2 RAM DETAILED ACTIVITIES, ANALYSES AND DOCUMENTS

Within this section, documents and activities necessary for defining and analysing reliability, availability and maintainability requirements are described.

5.2.1 RAM DESIGN RULES

It is planned to develop a series of rules, so-called RAM Design Rules as binding conditions for architectural design decisions if possible. Each of these RAM Design Rules will be mapped to one or more specific requirement(s). See section 3.6.

5.2.2 RAM HW-FMEA

It is recommended to create a RAM HW-FMEA, that aims to analyse availability and maintainability aspects in terms of failed HW components or failed sub-systems and their operational consequences. It also analyses diagnostic capabilities of the system in order to support preventive and corrective maintenance actions.

5.2.3 PRELIMINARY RELIABILITY ANALYSIS

Upon completion of the RAM Plan, it is recommended to create a preliminary Reliability Analysis document. This document must provide a rough estimation of the availability and reliability of the sub-systems described in chapter 3.

All relevant documentation shall be reviewed in order to define, at a preliminary level, the overall system structure and to recognise all the functional requirements, which may affect the RAM performance of the system.

5.2.4 RELIABILITY ANALYSIS

The Reliability Analysis is a key document, with focus on HW relevant aspects, describing reliability, availability and maintainability features.

Reliability aspects of SW have to be assured by following the rules of [EN 50126-1]. An analysis of SW reliability is not part of the Reliability Analysis where only permanent random HW faults and system failures will be analysed.

Failure rates for indoor components will be calculated

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To be able to make reliable statements and to demonstrate that the required RAM goals will be obtained, the following information about the hardware and the purpose of the system has to be collected:

- what kind of sub-systems / components will be used,
- quantity of each sub-system / component,
- Failure rates of all PBAs and devices used,
- MTBF and MTTR for each sub-system / component.

The Reliability Analysis shall point out measures, how availability can be reached according to the mission goals.

5.2.5 MAINTENANCE

To ensure a proper installation, operation and maintenance of the system, corresponding installation and maintenance documents have to be provided. MTTR and related terms have to be checked with respect to fulfilment of information stated in these documents.

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6. RAM DELIVERABLES

6.1 LIST OF RAM DELIVERABLE DOCUMENTS

This section summarizes the documents which are foreseen during RAM activities:

- RAM Plan (this document),
- RAM HW-FMEA (recommended),
- preliminary Reliability Analysis (recommended),
- Reliability Analysis (required).

Note: RAM documentation must be available according to the Document List and Review Matrix for engineering projects [DLRM].

6.2 RAM REQUIREMENTS

The RAM related customer requirements need to be checked, clarified if necessary and appropriately refined during the different development phases. If additional RAM related aspects are to be considered in order to fulfill RAM related customer requirements, they should comply with corresponding standards [EN 50126-1].

The solution element Digital Field Elements (DFE) are subject to the dependability requirements stated by the customer in [CRS_DSTW] and furthermore in [PH_RAMs].

The fulfillment of the following availability requirements must be shown in the Reliability Analysis.

Availability of ...	Class	Percentage	ID
FES	K2	99.995 %	02_124
FES – manual correction	K2	99.9979 %	02_127
FES – no operational impact	K2	99.7154 %	02_128
FES – 10 min automatic correction	K2	99.999%	02_129
FES – 2 min automatic correction	K2	99.9996 %	02_130
OC	K3	99.99 %	02_132
OC – manual correction	K3	99.9957 %	02_135
OC – no operational impact	K3	99.7154 %	02_136
OC – 10 min automatic correction	K3	99.999 %	02_137
OC – 2 min automatic correction	K3	99.9996 %	02_138

Table 7: Availability requirements Digital Field Elements

6.3 PERIODICAL RAM ACTIVITIES REPORT

The RAM Plan can be updated if necessary, in each phase of the product life cycle, described in [EN 50126-1].

Possible reasons are:

- Review comments of this document,

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- Hardware changes,
- Change of standards and procedures.

Each change of the RAM Plan results in a new edition. At each new edition, the reasons for change will be documented in section 1.5

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